Project Plan

Member Identification

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Project Overview – jy

1.1 Problem and Motivation

Our project proposes a multi-sensor context-aware campus exploration application set within the University of Melbourne. The application transforms campus navigation into a treasure-hunting experience in which users explore different locations to discover fictional lost relics and cultural treasures inspired by university landmarks.

Rather than unlocking a treasure solely when the user reaches a GPS coordinate, the application combines multiple contextual signals, including the user's location, movement or device stability, and facing direction. A treasure is revealed only when the required physical context is satisfied. This makes sensor interaction a core part of the exploration experience rather than an additional feature.

The project is designed to meaningfully integrate multiple sensors, Internet/cloud connectivity, responsive interaction, and a non-trivial mobile computing workflow.

1.2 Proposed Application

The proposed application, tentatively named Lost Treasures of UniMelb, allows users to explore the University of Melbourne campus and discover digital relics hidden around selected landmarks.

Users begin from a campus treasure map containing several treasure locations. Some information may initially remain hidden to encourage exploration. After selecting a treasure, the application retrieves its location, description, clue, and unlocking conditions from the cloud.

During exploration, the application continuously evaluates the user's physical context:

- GPS determines the user's current location, distance from the treasure, and target bearing.
- Accelerometer data determines whether the user is moving or sufficiently stable for scanning.
- Compass/orientation sensing determines the user's current heading and whether the device is aligned with the target direction.
- The Context Engine combines these signals to determine the current exploration state.

When the user enters the treasure's search area, the application transitions from general navigation into a Relic Scanner mode. The user may be asked to hold the phone steady and rotate towards the correct direction. Once the required location, stability, and directional conditions are satisfied, the treasure is revealed and added to the user's digital collection.

Cloud connectivity provides dynamic treasure information, while notifications can alert users to nearby or newly available treasures.

System Architecture –jy

2.1 Module Architecture

The system follows a modular architecture with six main functional components:

- Motion Detection Module
- Direction Module
- Location Module
- Context Engine
- User Interface
- Internet/Notification Module

The sensor modules independently process low-level physical-world information and expose simplified contextual states. The Context Engine acts as the central decision-making layer, while the UI translates context changes into navigation, scanning, and treasure-discovery feedback.

Remote treasure information is provided through the Internet layer and distributed to relevant application modules. This separation allows individual modules to be developed and tested independently while maintaining clear interfaces between team members.

2.2 Sensor and Data Flow

The application's main processing pipeline begins with physical user behaviour and ends with an adaptive interface response.

Raw sensor values are first handled by their corresponding modules. These modules transform low-level measurements into semantic states that are easier for the Context Engine to interpret.

For example, GPS determines whether the user is near the selected treasure, accelerometer processing determines whether the device is stable, and the Compass/Direction module determines whether the phone is facing the required direction.

The Context Engine combines these inputs and determines the current exploration state, such as NAVIGATING, SEARCHING, SCANNING, or TREASURE_REVEALED.

Rubric Coverage

Contribution Breakdown

1) Yunxiao Lu — Motion/Stability Detection

- Implement the accelerometer sensing module and collect real-time acceleration data from the Android sensor framework.
- Process and filter accelerometer readings to reduce sensor noise and improve motion-detection reliability.
- Develop logic to classify broad user movement states such as STATIONARY, WALKING, and MOVING.
- Implement device-stability detection to determine whether the phone is sufficiently still for the Relic Scanner stage.
- Define appropriate thresholds and expose structured MotionState and StabilityState outputs to the Context Engine.
- Test the module under different movement patterns and device-holding conditions, and assist with physical-device validation.
- Integrate the motion/stability outputs with the Context Engine and support end-to-end testing of the treasure-unlocking workflow.

2) Jiayi Yang — Compass and Direction Module

- Implement the Compass/Direction module using magnetometer data together with accelerometer or gravity information where required for device-orientation estimation.
- Calculate the device heading/azimuth and convert raw orientation readings into a stable direction representation.

- Integrate with the Location/GPS module to receive the target bearing for the selected treasure.
- Calculate and normalise the angular difference between the current heading and target bearing, including correct handling around the 0°/360° boundary.
- Apply smoothing and tolerance thresholds to reduce unstable compass behaviour caused by sensor noise or magnetic interference.
- Generate structured direction states such as TURN_LEFT, TURN_RIGHT, ALIGNED, and UNKNOWN for use by the Context Engine and reactive UI.
- Test directional behaviour on a physical Android device and integrate the module with the Location/GPS, Context Engine, and Relic Scanner workflow.

3) Fan Li – GPS/Location sensor developer

- Design and implement the GPS/location sensing module.
- Handle location permissions, availability states, and fallback behaviour.
- Tune update frequency for accuracy, responsiveness, and battery usage.
- Filter inaccurate or stale location readings.
- Test and prepare demo evidence for location-based behaviour.

4). Ruoning Ren—Code Integration and Context Engine Development

- Responsible for integrating the different modules made by team members into one mobile app and writing integration tests to make sure the whole process can work correctly on a real device.
- Help with version control and solve merge conflicts.
- Design and build a simple rule-based context engine that combines data from gyroscope, GPS, and motion detection to check what situation the user is currently in.
- Use conditions like distance to the target point, if user facing the correct direction, and if user is currently moving, to trigger notification or fitness task feedback

5) Zhuoer Liang – UI design and development

- **Interface and scenarios:** Design and implement the main interface framework and interaction scenarios for the app.
- **Navigation:** responsible for navigation and logical relationship among screens, ensuring a clear interaction flow
- **Information Visualization:** Selectively visualise information from sensors and the context engine, such as movement status, orientation, and whether the user is facing the correct direction, while handling different UI states.
- Interaction testing & user experience interview： test the interface on real Android devices and collect user feedback to refine the interface design.

6)Yunzhe Wang — Historical Treasure System and Content Integration

- Research and verify historical objects, photographs and events using official University of Melbourne sources, and define their connections with selected campus locations.
- Design the historical treasure experience, including the process from sensor-based unlocking to viewing the stylised 2D treasure and its related historical story.
- Define the treasure content structure, including treasure ID, location, unlock conditions, image reference, historical description and source information.
- Support the integration of treasure data with the cloud database and UI, and connect Context Engine states with the treasure reveal and historical information pages.
- Test the complete treasure-unlocking workflow on Android devices and prepare related visual materials and evidence for the final report and demonstration video.

Justification & Feasibility of Plan

I. Material – yunzhe

1. Report and Video

For Assignment 2, our group will prepare a final report and a demonstration video based on the marking rubric. The report will explain how our application meets each criterion and provides evidence for the main features.

The video will demonstrate the complete user experience on a real Android phone. The user will select a treasure location on the University of Melbourne campus, travel to the check-in area, and complete the required location, phone stability and direction conditions. After these conditions are met, the user can unlock a special digital treasure related to the building and learn about its history or a story that happened at that location. The video will also show that the sensor functions, cloud information, and user interface can work together as expected.

2. Compilation Screenshot

Before the final submission, we will build the completed application in Android Studio. A screenshot of the Android Studio Console will be included in the final report to clearly show that the submitted version is completed successfully.

3. Commit Log

Our group will use Git during the whole development process. Each member will use their own account to commit their work, and the commit messages will clearly describe the features, fixes or changes made. For Assignment 2, we will export the complete commit history with the authors, dates, and commit messages.

4. Itemised Contributions

The final submission will include a one-page itemised contribution list for all group members. It will describe the actual tasks completed by each person, including implementation, module integration, interface design, and testing. The final contribution list will be based on completed work and Git records.

II. Implementation

1.quality

To make sure our code meets these standards, our group will follow the same coding conventions throughout the project. This includes naming conventions for variables, functions, and files, so the code stays consistent no matter who writes it. We will also use the same code formatting style. Before merging a new feature into the main branch, we will also review each other’s code. During the review, we will check whether the code follows our naming and formatting conventions and whether it is easy for other team members to understand.

To keep the code readable and easy to understand, we will use clear and meaningful names for variables and functions instead of unclear names such as “data1” or “temp”. For each part of the project, we will add comments to explain the main logic and why we implemented it in a certain way. This will help other team members understand the code without having to ask the original author.

We believe these practices will help us keep the code organized and consistent throughout the project. They will also make it easier for all six team members to work on the same codebase and reduce confusion when combining different parts of the project.

2.sensors

We are planning to use multiple sensors, including gyroscope, GPS and motion sensor, to support our landmark-based app. Instead of using each sensor separately, we want to combine the data from different sensors to get a better understanding of user's location, direction and movement.

GPS will give the user's real-time location so the app can detect when they arrive at a landmark, and trigger a cultural pop-up, such as information about unimelb. Gyroscopes will help detect the direction the user is facing, which can be used to guide the user towards the next landmark. Motion detection will track user's movement, which can be used for navigation and for tracking whether the user completes the exercise.

We won't simply use one sensor API and read its raw values. Instead, we will combine information from different sensors to make the result more reliable. For example, different sensors can help reduce the limitations of using only one sensor. We will also do simple data smoothing to reduce noise from sensor readings, so the location detection and arrival trigger feel more accurate and reliable for the user. Also, these three sensors are commonly available on most mobile devices, so we don't need extra hardware.

3.connectivity

Our mobile app will use Internet data and cloud-based databases to support the treasure finding experience. For the users' location part, we plan to use Google Maps API to display the map, the users' location, and nearby landmarks at the University of Melbourne. The landmarks data will be stored in an online database to provide description, cultural information and images. When the users approach a treasured landmark, the app will display the corresponding landmarks' content.
Every time the users request Google Maps, it will be handled asynchronously in order not to block users' interfaces. For error handling, when the network is not stable, we will provide basic information to the users. Meanwhile, previously loaded landmark information can still be accessed because it is cached locally in the application.

We believe that this design meets the connectivity requirement because we will apply Google Maps API in our main functions including displaying the map, showing the location with the usage of GPS. And the cloud-based database is used for extensibility and scalability.

4.responsiveness

Our mobile app will show the users' location, movement, movement stage, and sensor without substantial delay. For the sensor usage, the GPS will continuously update the user's position on Google Maps, while the motion and accelerometer will show the direction and movement state of the users. When the user is close to the treasure target, the GPS will quickly give the signal and update the progress to show the target landmarks' information.

To reduce lag, sensor processing will be lightweight and easy to calculate. When designing the structure of the app, we should avoid unnecessary UI and request updates, what's more, Network request should be handled asynchronously to prevent blocking and heavy delay. We will also set a reasonable update frequency for the sensor and GPS update to keep performance-delay balance.

Android system can provide location and sensor callbacks correctly, so the plan is feasible technically. Our team believes these approaches can provide a quick and great experience for all users.

5.technical depth

We want the app to use advanced algorithms relating to localization and sensing. First, we plan to build a context engine that combines GPS, gyroscope, and accelerometer data using sensor fusion. This helps us better understand the user's real-time location and movement state, rather than rely on GPS alone.

For the localization side, we will implement a geofencing algorithm. Each landmark will have a defined radius on the map, and the app will check whether the user's real-time GPS location enters this area. When the user reaches a landmark, the app will trigger a culture pop-up and an exercise task. To make the result more reliable, the trigger will also check the motion sensor data so the app can confirm that the user is actually approaching and moving.

Our team believes this combination goes beyond basic sensor reading and will produce a more reliable result. The plan is realistic and achievable because the sensor APIs we are using are all standard and well-documented for mobile development, and both sensor fusion and geofencing are well-known techniques that don't require advanced mobile development knowledge. The modular development approach also means that each member can work independently, which reduces the risk of delay.

III. UI –zr

0. UI overview

The UI of this application will be designed around the theme of treasure hunting and exploration, adopting a light-hearted, engaging, and game-like visual style. The core interaction design will focus on creating an immersive, game-like narrative experience, while keeping the interface simple, clear, easy to learn, and easy to use.

The main pages will include: a **Home Page** for user login and displaying game content; a **Map Page** for presenting the campus map and treasure-related information; a **Task Page** for showing remaining tasks; a **User Settings Page** for viewing user information and current status;

as well as other supporting pages.

1. Appeal

We hope to use a consistent and interesting modern 2D illustration style, including the map, information hierarchy and visualisation, icons, and fonts.

The map will use a minimalist style and will only mainly mark places with treasures and mission points. The information on the map will be visualised in layers, and treasure locations will be highlighted in an important and interesting way. For icon, especially **each treasure image will also keep the main visual features of its historical prototype, such as the fossil specimen, and Atlas figure. This can allow each treasure to have its own historical identity.** The interface fonts will maintain a consistent style while choosing fonts that are more interesting but easy to read, avoiding fonts that are too fancy or too ordinary.

2. Guidelines Does it follow the UI guidelines of the operating system? 是否尊崇界面的设计规范

We will design the Android interaction guidelines by combining the characteristics of traditional mobile games and functional apps, and consider the design from three aspects: user habits, ergonomics, and system requirements.

In terms of **user habits,** we will use the commonly used Android **bottom navigation bar** to switch between the main pages, and use back navigation behaviours that Android users are familiar with. We will also provide visual feedback for states such as loading, errors, and success in ways that are consistent with Android conventions. Buttons, cards, dialogs, and other components will generally follow common Android design patterns, while more playful interactions will only be added to special interfaces, such as treasure-related pages.

In terms of **ergonomics**, we will ensure that buttons have sufficiently large **touch targets** for easy interaction. We will also maintain reasonable spacing, alignment, and visual hierarchy to make information easier for users to view and understand.

At the system level, we will use the standard Android **permission request** process, for example when requesting GPS or location permissions.

3. flow Is it clear and easy to access/find/activate the features of the app? 该程序是否各项清晰易用，方便用户查找和激活
After completing the required location and sensor conditions, the user will enter the treasure reveal page. The user will first see the stylised 2D treasure and can then swipe left to view its historical prototype photograph and related story. After reading the information, the user can return to the map and continue exploring other treasure

locations.（by yunzhe）

4. Language Are the labels and messages meaningful and appropriate? 标签和信息是否有意义和恰当

We aim to ensure that all textual and visual information presented through buttons and labels is concise, clear, and easy to understand.

For pop-up messages and instructional labels, we will use short and commonly understood expressions, avoiding technical terms such as *accelerometer* and *context engine*. Instead, complex technical concepts will be translated into language that fits familiar everyday experiences and the game context.

For game-related interfaces, the language will remain consistent with the overall narrative setting in order to maintain immersion, while still providing users with clear instructions on how to interact with the application correctly.

Prompt messages and state labels will also correspond closely to the current state of both the user and the device, so that the information provided is accurate and contextually appropriate.

5. Reactiveness Does the interface dynamically react to sensors or external data?

The interface will continuously reflect changes in the user's physical context, so that users can understand their current state and receive immediate feedback on what they should do next.

IV. innovation – fan

0. Innovation Overview

1. Novelty

app 的新功能或新组合，不只是普通 CRUD/app wrapper。
Unlike a normal campus map or check-in application, our app does not only display information when the user reaches a location. Each location unlocks a stylied digital treasure based on a real historical object, photograph, architectural feature or event related to the University of Melbourne. Users can explore both the creative treasure design and its historical background. （by yunzhe）

2. Surprise

说明它相对已有 app 有什么 unexpected extension。
The surprise comes from discovering hidden stories behind familiar campus locations. For example, Union Lawn can reveal a historical photograph of the former campus lake, while the South Lawn Underground Car Park can reveal a miniature reconstruction of an Atlas figure. Users first see the game-style treasure and can then swipe left to discover the real historical reference behind it. （by yuzhe）

3. Tech Knowledge

说明如何综合 mobile computing、cloud、sensing、HCI、data handling 等知识。

4. Cross-Disciplinary

说明借鉴了哪个领域，例如 urban planning, health, education, sustainability, transport, fitness, social behaviour。
The treasure design combines mobile computing and HCI with public history, cultural heritage, education, and visual design. Historical materials are redesigned as accessible digital collectables, while location-based interaction connects these stories with the real campus environment. （by yunzhe）

5. Innovation - Impact

说明它如何改善用户生活、任务或活动，并给出具体场景。
This design can help visitors, tourists, and new students understand the University of Melbourne in a faster and more engaging way. Instead of only reading long historical descriptions, users can discover short stories through visual treasures while visiting related locations. The treasure-hunting experience may also encourage users to explore more parts of the campus. （by yunzhe）

~~Feasibility of Plan(?可写在 imetmplement 中）~~

~~Ai use~~

Each treasure will be inspired by a real historical object, photograph, architectural feature or event related to a University of Melbourne location. The treasure itself will be presented as a stylized 2D design, while the historical reference and story will be shown on a separate information page. （by yunzhe）

Campus Location	Digital Treasure	Historical Connection
Union Lawn	The Lost Lake Photograph	A historical photograph card based on the former campus lake, which was later filled in and converted into Union Lawn.
Wilson Hall	The Surviving Stone Rosette	A stylised stone rosette based on architectural elements reused when the current Wilson Hall replaced the original hall destroyed by fire in 1952.
Old Quad	Miniature Fossil Specimen	A miniature fossil based on the drawings of Mesozoic ferns fossilised in sandstone, which later inspired the stained-glass installation at Old Quad.
South Lawn Underground Car Park	Miniature Atlas Figure	A reconstructed miniature model based on the two Atlas figures located at the western entrance of the car park. These figures originally came from the demolished Colonial Bank building.
System Garden	Miniature Glasshouse	A miniature reconstruction of the former octagonal glasshouse. The remaining central tower was originally connected to the conservatory before the glasshouse was dismantled in 1916.
Grainger Museum	Miniature Kangaroo-Pouch Tone-Tool	A miniature model based on the experimental music machine created by Percy Grainger and Burnett Cross in 1952.

The treasures in our application will be digital collectables inspired by verified historical photographs, objects, architectural features and events related to the University of Melbourne. Although some treasures are designed as miniature models, they will be presented as stylised 2D illustrations in the current version of the app. Users can swipe left from the treasure page to view its historical reference and related story.

1. Union Lawn — The Lost Lake Photograph

Union Lawn was previously the location of a campus lake created after a creek was dammed in 1861. The lake was used for activities such as boat races, yabbying and mock battles before being gradually filled during the 1930s. The treasure will be designed as a stylised historical photograph card showing the former lake.

2. Wilson Hall — The Surviving Stone Rosette

The original Wilson Hall opened in 1882 but was destroyed by fire in 1952. The current Wilson Hall opened in 1956 and reused some parts of the original building, including stone rosettes. The treasure will be a stylised stone rosette representing the connection between the original and present Wilson Hall.

3. Old Quad — The Miniature Fossil Specimen

The stained-glass installation at Old Quad was inspired by two historical drawings of Mesozoic fern fossils preserved in sandstone. The drawings were created by Arthur Bartholomew and Ludwig Becker and later survived as lithographs. The treasure will be a miniature fossil specimen based on the fossil prototype rather than the stained glass itself.

4. South Lawn Underground Car Park — The Reconstructed Miniature Atlas Figure

The western entrance of the South Lawn Underground Car Park contains two Atlas figures from the demolished Colonial Bank building in Melbourne. The treasure will be a reconstructed miniature model of one Atlas figure, helping users notice this historical architectural feature that may otherwise be missed.

5. System Garden — The Miniature Glasshouse Model

The tower in the centre of the System Garden was originally connected to a unique octagonal glasshouse. The glasshouse was dismantled in 1916, while the tower remained. The treasure will be a miniature reconstruction of the complete glasshouse, allowing users to imagine the original appearance of the garden.

6. Grainger Museum — The Miniature Kangaroo-Pouch Tone-Tool Model

The Kangaroo-pouch Tone-tool was created by Percy Grainger and Burnett Cross in 1952 as part of Grainger's Free Music experiments. It used cut paper reels, mechanical components and electronic oscillators to control musical pitch. The treasure will be a miniature model of this experimental music machine.

by yunzhe

Note: The treasure illustrations shown above were generated using AI and are provided only as visual references for the early prototype. They are not final designs and should not be considered exact reconstructions of the original historical objects.